

Sparsh Kumar

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Summary: I'm a CS junior at San Jose State (B.S., December 2027) who built interest in software development through teaching and competing. My work runs from embedded firmware to ML data pipelines, financial tooling, and full-stack apps. Beyond the code, I founded a financial literacy program for 50+ students, led a CS club, and placed in coding and math competitions. Hope to continue learning.

Education:

San Jose State University B.S. in Computer Science | Expected Graduation December 2027

Sierra College A.S. in Computer Science, Mathematics, and Physics | Graduated May 2025

Certifications: [AWS Cloud Practitioner Foundational](#)

Experience:

Research Assistant (Software Developer) – De Novo Protein Research Lab, UC Riverside

Remote | May 2025 - March 2026

- Wrote the lab's first Python parser for RFdiffusion .trb metadata, replacing manual file-by-file extraction across batches of about a 100 PDB/TRB structure files.
- Extended it into an **automated Python/Bash pipeline** that pulls outputs from the lab's compute cluster and re-extracts metadata after each RFdiffusion round, scaling from 5 seed designs to 125 across three rounds of 5x expansion.
- Standardized JSON and metadata I/O so outputs** feed directly into AlphaFold, ProteinMPNN, LigandMPNN, and PyRosetta.
- Built Biopython analysis and visualization scripts with a team of biologists and chemists, and debugged legacy and third-party code.
- Cut processing time from hours to minutes by **automating** data entry, type conversion, and file pipelining previously done by hand.

Founder & Instructor - Financially Free

Rocklin, CA | May 2024 - April 2025

- Founded and led a financial literacy program across the middle school district, teaching 50+ students core concepts in debt, credit, interest, and investing.
- Built a stock-market simulator with simulated price movement (live trades, portfolio tracking, real-time P&L) and a credit-score model reflecting payment history, utilization, and borrowing behavior.

President - Sierra College Computer Science and Math Club

Rocklin, CA | January 2024 - March 2025

- Administered the Club website and Game development projects. Used **GitHub, Java, HTML, CSS, JavaScript, and Unity**.
- Planned and executed 3 coding competitions for 40+ competitors. Authored **problems/solutions** for the competition.
- Hosted an orientation event for 50+ new students. Planned venue, food, and brought in professors/professionals to share insight.

Projects:

RIM | Basketball Social App

- Built a social app for pickup basketball groups: friends log stats across games they play together, track head-to-head records(leaderboard), and plan hooping sessions with time, location, attendance voting, and more.
- Synced RSVPs across devices in real time via Supabase Realtime channels over WebSockets, subscribing to Postgres change events.
- Built stat and leaderboard views in TypeScript with typed API models and virtualized lists over a group's full game history.
- Designed the normalized PostgreSQL schema and **SQL aggregation queries** behind the leaderboards, with indexed lookup paths.

Financial Research Tool

- Built a **RAG pipeline** that chunks and embeds SEC filings and earnings transcripts into a **ChromaDB** vector store, with **LangChain** retrieval feeding the Claude API.
- Generated answers grounded in retrieved source sections, surfacing risk factors and guidance changes without reading full 10-Ks.
- Developed a Streamlit interface unifying document Q&A and portfolio tracking in one workflow.

Arduino Project | Embedded Security System and Autonomous Vehicle

- Built a multi-sensor intrusion detection system in C++ and Assembly (motion, IR, light, sound), using low-level I/O control to optimize performance and logging activity patterns to flag anomalies.
- Programmed an autonomous vehicle in C++ to navigate obstacles using ultrasonic distance sensing.

Skills:

Languages: Python, C++, Java, JavaScript, SQL, Rust, Assembly

ML & Data: PyTorch, Pandas, NumPy, Biopython, LangChain, Chroma, RAG

Infrastructure & Tools: FastAPI, REST APIs, Docker, Git/GitHub, Kubernetes, Linux, Bash, AWS, PostgreSQL/Supabase, Arduino

Competitions:

- Top 10 at UC Davis Coding Competition** | Used **Python**
- Top 5 at Sierra College AMATYC Math contest**
- Top 3 at Sierra College Coding Exhibition** | Used **C++**

Davis, CA | May 2024
Rocklin, CA | March 2024
Rocklin, CA | November 2023